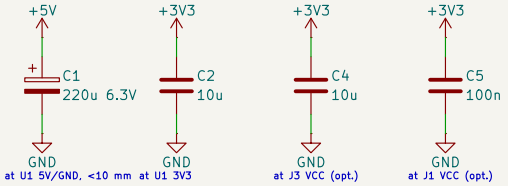
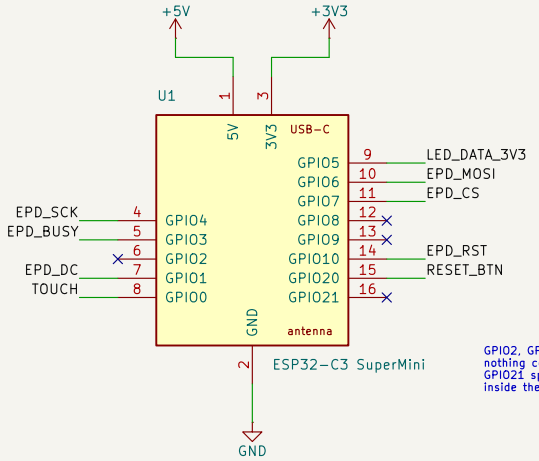
Connected Heart — carrier PCB rev A

ESP32-C3 SuperMini carrier: 5 V and 3.3 V come from the SuperMini's own USB-C and LDO. No battery, no regulator, no second USB port (ADR-0005).
Off-board modules on harnesses: J1 TTP223 touch, J2 3x WS2812B, J3 2.13" e-paper (SSD1680). Pin numbers on U1 are the SuperMini's physical order, top view, USB-C up.

Decoupling (place each one at the pin named under it)

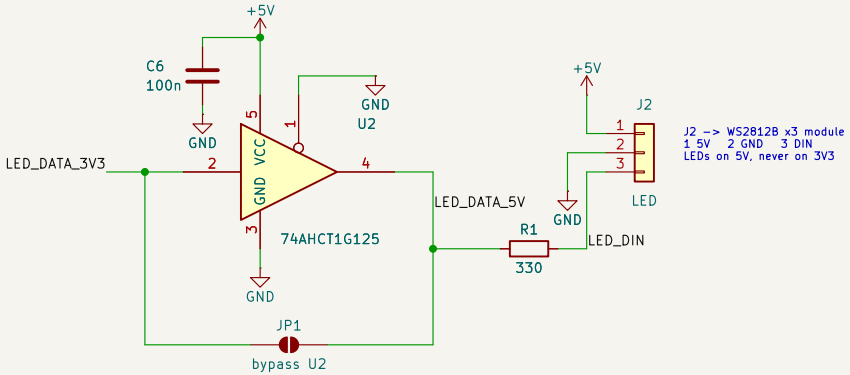


USB-C on the SuperMini is the only port:
power + flashing. 5V pin = VBUS
pass-through; 3V3 = onboard LDO (~500 mA).



GPIO2, GPIO8, GPIO9 are strapping pins:
nothing connected on the carrier.
GPIO21 spare. GPIO18/19 = native USB,
inside the module.

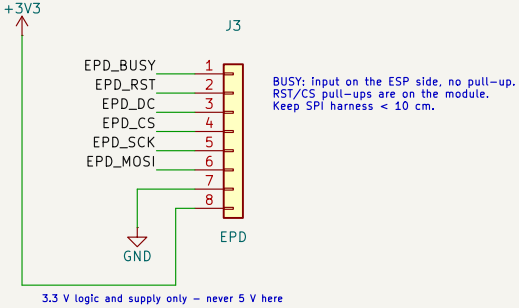
LED data: 3.3 V GPIO5 -> 5 V level shifter -> 330 R -> J2. Fit U2 OR close JP1, never both.



J2 -> WS2812B x3 module
1 5V 2 GND 3 DIN
LEDs on 5V, never on 3V3

WS2812B needs V_IH >= 3.5 V at 5 V; 3.3 V drive usually works but is out of spec.
Fit U2 and C6, leave JP1 open. JP1 is a bypass only: fit U2 or JP1, never both.

J3: e-paper module cable (Waveshare 2.13" order — verify on your module)

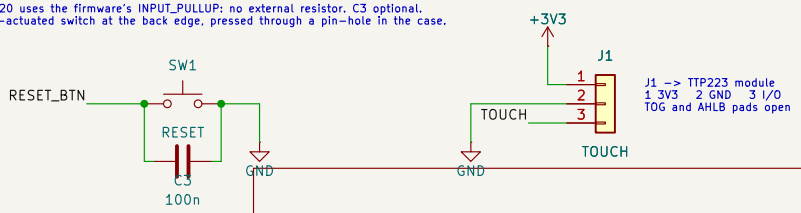


BUSY: input on the ESP side, no pull-up.
RST/CS pull-ups are on the module.
Keep SPI harness < 10 cm.

J1: TTP223 touch (active-high, momentary)

SW1: factory reset

GPIO20 uses the firmware's INPUT_PULLUP: no external resistor. C3 optional.
Side-actuated switch at the back edge, pressed through a pin-hole in the case.



J1 -> TTP223 module
1 3V3 2 GND 3 I/O
TOG and AHLB pads open

Mounting: 4x M2.5, corners, no copper connection (case screws must not tie into the ground pour)



Pins match firmware/src/Core/Pins.h

Connected Heart

Sheet: /
File: connected-heart-carrier.kicad_sch

Title: Connected Heart — carrier PCB

Size: A4	Date: 2026-09-15	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1